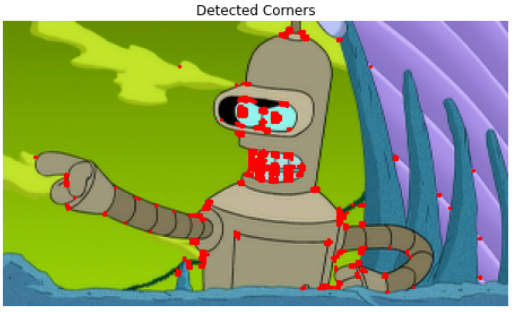
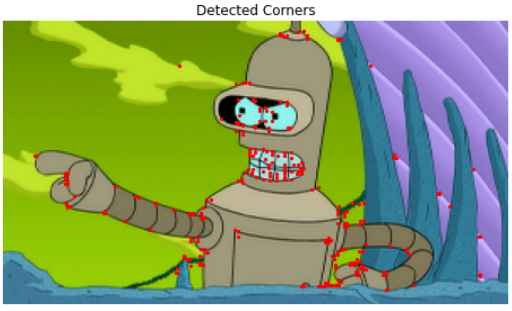
Classical_CV

1. Harris Corner Detection
    1. Gradient computation
        1. Compute the gradients I_x and I_y using the Sobel operators:
        2. $I_x = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}$
        3. Convolve the grayscale image with I_x and I_y to obtain the gradient images I_x and I_y.
    2. Structure Tensor calculation
        1. Compute the elements of the structure tensor:
        2. I_x^2, I_y^2, and $I_x \cdot I_y$
        3. These computations involve element-wise multiplication of I_z and I_y with themselves.
    3. Harris response calculation
        1. For each pixel (i, j), compute the elements of the structure tensor within a local window of size $N \times N$. Compute the sums of squares of I_x and I_y within the window to obtain S_{xy}. Compute the determinant and trace of structure tensor:
        2. $\det(M) = S_{xx} \times S_{yy} - S_{xy}^2$, $\mathrm{Tr}(M) = S_{xx} + S_{yy}$
        3. Compute the Harris response (R) using the formula:
        4. $R = \det(M) - k \times (\mathrm{Tr}(M))^2$
        5. where k is an empirically chosen constant.
    4. Thresholding and Non-maximum Suppression
        1. Apply a threshold to R to identify potential corner candidates. Perform non-maximum suppression to retain only local maxima as corner points.
    5. Visualization
        1. Plot the original image with detected corner points marked.
    6. Before non-max suppression



    7. After non-max suppression



    8. Code:

```
# -*- coding: utf-8 -*-
"""
Created on Tue Feb 17 22:10:18 2026

@author: reina
"""

from scipy.ndimage import convolve
from skimage import color
from skimage import io
import numpy as np
import cv2
```

```
import matplotlib.pyplot as plt

def harris_corner_detection(image, threshold=0.01, window_size=3, k=0.04):
    # Convert the image to grayscale
    gray = color.rgb2gray(image)

    # Compute gradients using Sobel operators
    Ix = convolve(gray, np.array([[-1, 0, 1], [-2, 0, 2], [-1, 0, 1]]))
    Iy = convolve(gray, np.array([[-1, -2, -1], [0, 0, 0], [1, 2, 1]]))

    # Compute elements of the structure tensor
    Ix2 = Ix * Ix
    Iy2 = Iy * Iy
    Ixy = Ix * Iy

    # Compute Harris response
    height, width = gray.shape
    R = np.zeros((height, width))
    offset = window_size // 2
    for i in range(offset, height - offset):
        for j in range(offset, width - offset):
            Sxx = np.sum(Ix2[i - offset:i + offset + 1, j - offset:j + offset + 1])
            Syy = np.sum(Iy2[i - offset:i + offset + 1, j - offset:j + offset + 1])
            Sxy = np.sum(Ixy[i - offset:i + offset + 1, j - offset:j + offset + 1])
            det = Sxx * Syy - Sxy**2
            trace = Sxx + Syy
            R[i, j] = det - k * trace**2

    # Thresholding
    corners = np.zeros_like(R)
    corners[R > threshold * np.max(R)] = 1
    # Perform non-maximum suppression
    corners = non_maximum_suppression(corners)

    return corners

def non_maximum_suppression(corners):
    # Define 8-neighbor offsets
    offsets = [(i, j) for i in [-1, 0, 1] for j in [-1, 0, 1] if (i != 0 or j != 0)]

    # Create a copy of the corners matrix
    suppressed_corners = corners.copy()

    # Iterate over each pixel in the corners matrix
    height, width = corners.shape
    for i in range(height):
        for j in range(width):
            # Check if the current pixel is a corner
            if corners[i, j] > 0:
                # Calculate the sum of the values of the neighboring pixels
                neighbor_sum = 0
                for offset_i, offset_j in offsets:
                    ni, nj = i + offset_i, j + offset_j
                    if ni >= 0 and ni < height and nj >= 0 and nj < width:
                        neighbor_sum += corners[ni, nj]

                # If the sum of neighboring pixels is less than 4, keep the current pixel as a corner
                if neighbor_sum < 4:
                    suppressed_corners[i, j] = 1
                else:
                    suppressed_corners[i, j] = 0

    return suppressed_corners

# Load the example image
img = cv2.imread('bender.png')
img = cv2.resize(img, (300, 169), interpolation=cv2.INTER_CUBIC)
#plt.imshow(img, cmap='gray')

# Perform Harris corner detection
corners = harris_corner_detection(img)

# Visualize the detected corners
plt.figure(figsize=(8, 6))
plt.imshow(img)
plt.scatter(np.nonzero(corners)[1], np.nonzero(corners)[0], color='r', s=5)
plt.title('Detected Corners')
plt.axis('off')
plt.savefig('harris.png')
plt.show()
```

2. SIFT
    1. People say 'SIFT' but there are two parts to SIFT [1].
        1. an interest point detector.
        2. a region descriptor.
            1. They are independent, many people use the region descriptor without looking for the points.
    2. SIFT summary [2]:

1. Extract a 16x16 patch around detected keypoint.
2. Compute the gradients and apply a Gaussian weighting function.
3. Divide the window into a 4x4 grid of cells.
4. Compute gradient direction histograms over 8 directions in each cell.
5. Concatenate the histograms over 8 directions in each cell.
6. Concatenate the histograms to obtain 128 dimensional feature vector.
7. Normalize to unit length.

	Rotation Invariant	Scale Invariant	Affine Invariant	Repeatability	Localization Accuracy	Robustness	Efficiency
Haris	x			+++	+++	++	++
Shi-Tomasi	x			+++	+++	++	++
FAST	x	x		++	++	++	++++
SIFT	x	x	x	+++	++	+++	+
SURF	x	x	x	+++	++	++	++

Image Courtesy of Davide Scaramuzza et. al 2012.

8. Local features: main components [3]
    1. Detection: Identify the interest points.
    2. Description: Extract vector feature descriptor surrounding each interest point.
    3. Matching: Determine correspondence between descriptors in two views.

3. ORB
    1. FAST (detector), BRIEF (descriptor)
    2. ORB (Oriented FAST and Rotated BRIEF) combines FAST and BRIEF, adds some orientation to the FAST and adds some rotation to the BRIEF, so this will make it more robust.
4. Optical flow
    1. Motivation: we want to match features between two frames, however we don't want to perform feature matching from one view to another view. Can we just first detect some of the key points or interest points corners, edges, etc, and then use some algorithm to know the correspondences in the next image without searching for the correspondences. Because this searching for correspondences consumes a lot of time. A more efficient way is to use algorithm to compute the locations of these correspondences in the next frame, thus we can form correspondences in this approach. This algorithm is called Lucas-Kanade algorithm (LK algorithm).
    2. Estimate the motion of each pixel between two consecutive image frames, based on assumptions:
        1. brightness consistency
        2. small motion
    3. Can we use accelerometer instead? yes, however acceleration needs to be integrated twice to obtain position.
    4. with optical flow we have one less integration error.
    5. we want to estimate pixel motion from image I(x, y, t) to image I(x, y, t+1).
    6. Optical flow: *Apparent* motion of objects.
    7. examples where:
        1. There is no optical flow, but there is relative motion (between camera and environment): Could happen in textureless environment.
        2. There is no relative motion, but there is optical flow: changing light source or intensity, the sun or some other source of light, the camera is fixed relative to the environment but you can see apparent change of the environment.
    8. Apparent motion could be due to different factors:
        1. Motion of object in camera's view.
        2. Motion of camera itself.
        3. Lighting changes.
    9. Birds rely on trick of the eye: optical flow. Optical flow cues in birds. Birds tend to keep similar optical flow on the left and right eyes . Birds use optical flow to fine-tune their navigation when manuevering through narrow spaces at high

speeds.

References:

1. CSE576 Udub Linda Shapiro
2. Trym Vegard Haavardsholm, UNIK4690
3. CSIE 5429 3D-DLCV, 3D Computer Vision with Deep Learning Applications, National Taiwan University, Spring 2021.
4. Lecture 18, Princeton, Introduction to Robotics, Optical flow.